



Ankit Kumar

Computational Chemist | Drug Discovery & Machine Learning

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🌐 LinkedIn 🐙 GitHub 🌐 Portfolio 🏠 Google Scholar

Professional Profile

Computational drug discovery scientist (Ph.D. Chemistry) with over seven years of research experience, combining hands-on machine learning and cheminformatics (RDKit, scikit-learn, PyTorch) with structure-based design (Schrödinger Maestro, LiveDesign) and molecular dynamics (GROMACS). Builds and validates QSAR and deep-learning models for potency prediction, including scaffold-based validation, conformal uncertainty estimation, and applicability-domain assessment, and deploys them as production services (FastAPI, Streamlit, Docker).

Skills

Machine Learning & AI: Machine Learning Algorithms, Deep Learning, Graph Neural Networks

Cheminformatics & QSAR: RDKit, Molecular Fingerprints & Descriptors, QSAR Modeling, Chemprop

Computer-Aided Drug Design (CADD): Molecular Docking, Virtual Screening, Free Energy Perturbation (FEP), Molecular Dynamics (GROMACS)

Programming & MLOps: Python, Scikit-Learn, PyTorch, SQL, Git, Docker, FastAPI, Streamlit, Weights & Biases, GCP

Data Sources & Platforms: ChEMBL, PubChem, PDB, Schrödinger Maestro, LiveDesign

Experimental & Analytical: Organic Synthesis, Process Chemistry, Scale-Up, NMR, HR-MS, HPLC, GC-MS, X-ray Crystallography

Professional Experience

AI Trainer

Mar 2025–Present

Outlier.ai | Troisdorf, Germany (Remote)

- Create and curate high-quality datasets and evaluation frameworks for STEM-focused AI models and drug discovery agents
- Review, edit, and improve work submitted by peers to align with client requirements and quality standards
- Provide feedback on chemical data accuracy and scientific validity for AI training datasets

Career Transition: AI & Machine Learning in Chemistry

Apr 2024–Feb 2025

Self-Directed Learning | Troisdorf, Germany

- Self-study in machine learning fundamentals, deep learning, and AI applications in chemistry

Data Science Trainee

Nov 2023–Mar 2024

neuefische GmbH — School for Digital Talent | Hamburg, Germany (Remote)

- Intensive bootcamp in ML algorithms, statistical modeling, and end-to-end data science workflows

Full-Time Parenting

Jul 2022–Oct 2023

Troisdorf, Germany

Postdoctoral Research Associate

Dec 2020–Jun 2022

Technical University Dresden | Dresden, Germany

- Synthesized and analyzed carbon-based materials such as graphene nanoribbons and spin chains
- Developed scalable methodologies for hetero-structure graphene nanoribbons for optoelectronic applications

Senior Research Associate

Jan 2020–Oct 2020

Neogen Chemicals Limited | Vadodara, India

- Developed scalable methods for mass production of pharmaceutical intermediates

- Achieved 10–15% yield improvements, reducing production costs and enhancing profitability

Education

Doctor of Philosophy (Ph.D.) in Chemistry

Indian Institute of Technology Bombay, Mumbai, India | Jul 2014–Dec 2019

- Developed polyaromatic hydrocarbon based porphyrinoid macrocycles and metal complexes
- Strong experience with NMR, HR-MS, UV–Vis, fluorescence, electrochemistry, and X-ray crystallography

Master of Science (M.Sc.) in Chemistry

CCS University Meerut, India | 2010–2012

Selected Projects

EGFR pIC₅₀ Prediction: QSAR & Deep Learning Models for Kinase Inhibitor Potency — [GitHub](#) | [Live Demo](#)

- Curated 10,502 EGFR bioactivities from ChEMBL (ChEMBL203) with RDKit standardization and Bemis-Murcko scaffold splitting for realistic generalization testing
- Trained and compared classical ML (Random Forest, XGBoost, LightGBM on ECFP4 + RDKit descriptors), a message-passing GNN, Chemprop D-MPNN, and a CheMeleon fine-tune; Random Forest performed best (RMSE 0.889, Spearman ρ 0.73)
- Shipped every prediction with a conformal uncertainty interval and applicability-domain flag; deployed as a Streamlit app and FastAPI service in Docker

Tech: Python, RDKit, Scikit-Learn, Deep Learning, PyTorch Geometric, Chemprop, FastAPI, Streamlit, Docker, GCP

AqSolPrediction: Aqueous Solubility Prediction — [GitHub](#) | [Live Demo](#)

- Built ML regression pipeline on AqSolDB (~10k compounds) using RDKit molecular descriptors and Morgan Fingerprints, achieving test $R^2 = 0.88$
- Deployed interactive web app on Hugging Face Spaces with real-time SMILES input and prediction

Tech: Python, RDKit, Hugging Face, Docker, FastAPI

Imatinib Resistance in BCR-ABL1: All-Atom MD of the T315I Gatekeeper Mutation — [GitHub](#)

- Designed a 2x2 panel (imatinib/ponatinib × wild-type/T315I) and ran 12 × 100 ns GROMACS simulations (CHARMM27, explicit solvent, in-silico mutagenesis) on a GCP L4 GPU
- Reproduced the clinical resistance mechanism: T315I collapsed imatinib's gatekeeper contact from 97% to 53% occupancy and cost 0.6 H-bonds/frame, while ponatinib was unaffected
- Built the RMSD/RMSF, H-bond, and pocket-contact analysis pipeline in Python

Tech: GROMACS, CHARMM27, ChimeraX, PyMOL, VMD, Python, NumPy, Matplotlib, GCP

Certifications & Publications

- Schrödinger – Applications of Free Energy Calculations in Modern Drug Hunting (2026)
- Schrödinger – Virtual Screening with Integrated Physics and Machine Learning (2026)
- Schrödinger – Target Enablement, Preparation, and Validation (2026)
- Schrödinger – Designing Quality Ligand Libraries (2026)
- Schrödinger – Introduction to Molecular Modeling in Drug Discovery
- MD Simulations & Computational Biophysics (GROMACS, VMD, ChimeraX) – BDGLifeSciences
- NVIDIA Certified Associate – Generative AI & LLMs (2024)
- Author of 13 peer-reviewed scientific publications in international journals.

Languages & Additional Information

German: B1 Certified **English:** Full Professional Proficiency **Hindi:** Native

Work Authorization: Valid German residence and work permit; open to relocation